Heim's Unified Mass Formula

Consolidated Reference (1982 Basis & 1989 Refinements)

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Abstract

This document consolidates the algorithmic determination of elementary particle masses according to Burkhard Heim. It integrates the invariant definitions from the 1982 DESY programming manuscript with the critical structural corrections, lost bracket restorations, and parameter updates (neutrino masses, lifetimes) presented in the 1989 MBB/DASA report.

1 Fundamental Invariants and Metric Tensor

The mass spectrum is determined by the geometric configuration of the internal dimensions ($R_6 \to R_4 \to R_3$). The primary invariants defining a state (multiplet) are:

- **Configuration Number (k)**: Metric index. $k = 1$ (Mesons), $k = 2$ (Baryons). ($k = 0$ implies no ponderable mass).
- **Time Helicity (ε)**: $\varepsilon = \pm 1$. Differentiates between structure and anti-structure.
- **Isospin Parameter (P)**: Related to Isospin I by $I = P + 1$.
- **Spin Parameter (Q)**: Related to Spin J by $Q = 2J$.
- **Doublet Index (K)**: Distinguishes multiple doublets (0 or 1).

1.1 1989 Correction to Charge and Strangeness

The 1989 report enforces a scaling of the structural distributor C (Strangeness) and updates the charge quantum number definition.

1. Distributor Scaling: The parameter C defined in 1982 Eq.(I) must be scaled by the configuration number:

$$C_{1989} = \frac{C_{1982}}{k} = \frac{2(P\varepsilon_P + Q\varepsilon_Q)(k - 1 + K)}{k(1 + K)} \tag{1}$$

2. Charge Quantum Number (q_x): The explicit expression for the charge state x ($0 \leq x \leq P$) is updated to:

$$q_x = \frac{1}{2}\left[(P - 2x + 2)[1 - KQ(2 - k)] + \varepsilon[k - 1 - (1 + k)Q(2 - k)] + C_{1989}\right] \tag{2}$$

The absolute charge magnitude is $q = |q_x|$.

3. Angle Definition (α_Q): The angle defining the helicity projection is corrected:

$$\alpha_Q = \pi Q \left[Q + \binom{P}{2}\right] \tag{3}$$

2 The Consolidated Mass Operator

The mass M of a state is generated by the cyclic occupation of the 4-fold metric contours. The 1989 formalism modifies the 1982 linear summation into a structure including the fine-structure terms $\alpha_\pm$.

2.1 The Master Equation (1989)

$$M = \mu\alpha_+ \left[(G + S + F + \Phi) + 4q\alpha_-\right] \tag{4}$$

Components:

- μ: The mass element derived from general relativity and quantum constants (Eq. VI, 1982).
- G, S: Structural metric terms (equivalent to $\underline{G}, K$ in 1982, but using indices n, m, p).
- F: The functional interaction term (updated H).
- Φ: The macroscopic coupling term.

2.2 Fine Structure Constants

The 1989 report introduces specific forms for the coupling constants, derived from the geometric auxiliary $\eta = \pi(\pi^4 + 4)^{1/4}$:

$$\alpha_+ = \frac{\sqrt[6]{\eta}}{\eta^2} \quad (\approx \text{inverse of mass scale}) \tag{5}$$

$$\alpha_- = (\alpha_+ + 1)\eta - 1 \tag{6}$$

2.3 Interaction Functions (F and Φ)

Using the 1989 notation for zone occupation numbers: n (Zone 1), m (Zone 2), p (Zone 3), σ (Zone 4).

The Term F:

$$\begin{aligned} F = {} & 2nQ_n[1 + 3(n + Q_n + nQ_n) + 2(n^2 + Q_n^2)] \\ & + 6mQ_m(1 + m + Q_m)N_2 \\ & + 2pQ_pN_3 + \phi(p, \sigma) \cdot \delta(N) \end{aligned} \tag{7}$$

Note: Q_n, Q_m, Q_p correspond to the metric invariants Q_1, Q_2, Q_3 from the 1982 manuscript.

The Term Φ:

$$\Phi = P(-1)^{P+Q}(P + Q)N_5 + Q(P + 1)N_6 \tag{8}$$

The Self-Coupling $\phi(p, \sigma)$: Crucial for lifetime calculations (appears only in basic states where $N = 0 \implies \delta(0) = 1$):

$$\begin{aligned} \phi = {} & \frac{N_4p^2}{1 + p^2} \frac{\sigma + Q_\sigma}{\sqrt{1 + \sigma^2}} \left(\sqrt[4]{2} - 4BUW_{N=0}^{-1}\right) \\ & + P(P - 2)^2 \left(1 + \frac{K(1 - q)}{2\alpha\vartheta}\right) \left(\frac{\pi}{e}\right)^2 \sqrt{\eta_{12}}(Q_m - Q_n) \\ & - (P + 1)\binom{Q}{3}/\alpha \end{aligned} \tag{9}$$

3 Selection Rules and Resonance Barriers

The physical realization of a particle depends on solving the implicit "Selection Rule."

3.1 Occupational Limits (The "Grid")

The occupation numbers n, m, p, σ are bounded by barriers L determined by the previous zone's saturation.

$$-Q_n \leq n \leq L_{(n)} \tag{10}$$
$$-Q_m \leq m \leq L_{(m)}(n) \tag{11}$$
$$-Q_p \leq p \leq L_{(p)}(m) \tag{12}$$
$$-Q_\sigma \leq \sigma \leq L_{(\sigma)}(p) \tag{13}$$

The calculation of L values requires the root-finding of the cubic saturation equations (e.g., 1989 Eq. after B13).

3.2 The Eigenvalue Equation ($W_{\nu x}$)

The allowed states are roots of the unified interaction equation:

$$\sum \cdots + \exp\left[\frac{1-2k}{3Q_\sigma}(\sigma + Q_\sigma)\right] = W_{\nu x}[1 + f(N)] \tag{14}$$

The 1989 Base Potential $W_{N=0}$ is explicitly:

$$W_{N=0} = Ae^x(1-\eta)^L + (P-Q)\left(1 - \binom{P}{2}\right)\left(1 - \binom{Q}{3}\right)(1-\sqrt{\eta})^2\sqrt{2} \tag{15}$$

where A, x, L are complex combinatorial functions of the invariants k, P, Q, H, g (See 1989 "Table VI/chapter G" definitions).

4 Implementation Constants

Metric Invariants ($s = k^2 + 1$):

$$Q_n = 3 \cdot 2^{s-2}$$
$$Q_m = 2^s - 1$$
$$Q_p = 2^s + 2(-1)^k$$
$$Q_\sigma = 2^{s-1} - 1$$

Physical Constants (1982/1989):

- $\hbar = 1.0545887 \times 10^{-34}$ J s
- $c = 2.99792458 \times 10^8$ m/s
- $\alpha^{-1} \approx 137.036$ (Calculated endogenously via η functions)